

Aim

To create a simple Android application using Kotlin and understand Activity, layout, and displaying output on the screen.

Procedure

1. Open Android Studio.
2. Create a new project:
 - o Select **Empty Activity**
 - o Language: **Kotlin**
 - o Minimum SDK: **API 21 or above**
3. Open res/layout/activity_main.xml.
4. Add a **TextView** and **Button**.
5. Open MainActivity.kt.
6. Use findViewById to access UI components.
7. Set a click listener for the button.
8. Change TextView text on button click.
9. Run the app on emulator/device.
10. Observe the output.

Code**activity_main.xml**

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">
```

```
    <TextView
        android:id="@+id/tvMessage"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello, Android!"
        android:textSize="22sp"/>
```

```
    <Button
        android:id="@+id/btnClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
        android:text="Click Me"/>
    </LinearLayout>
```

MainActivity.kt

```
package com.example.myfirstapp

import android.os.Bundle
import android.widget.Button
import android.widget.TextView
import androidx.appcompat.app.AppCompatActivity

class MainActivity : AppCompatActivity() {

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        val tvMessage = findViewById<TextView>(R.id.tvMessage)
        val btnClick = findViewById<Button>(R.id.btnClick)

        btnClick.setOnClickListener {
            tvMessage.text = "Button Clicked! Welcome to Android."
        }
    }
}
```

Result

The application successfully displays:

Initial output:

Hello, Android!

After button click:

Button Clicked! Welcome to Android.

